# BayesINE5.
# Modelos dinámicos lineales para series temporales

## Intro a Estadística Bayesiana

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DataLab

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ICMAT-CSIC

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# Chapter objectives

# Objectives

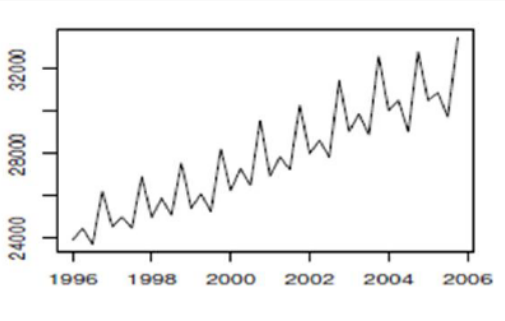
- Overview key concepts in Bayesian time series modeling with the aid of Dynamic Linear Models
- Prediction, Prediction, Prediction
- The lab reinforces concepts operationally

# Approaches to time series models

# Problem setup

Problem faced: Forecasting the values of a time series of interest

An example: *We have available quarterly consumptions of gas in Spain. We need to forecast consumption over the next year to plan production.*




Numerical example in Lab. Here, concepts and methods.

Check summary in

`https://www.math.unm.edu/~ghuerta/tseries/dlmch2.pdf`

# Approaches to time series

Three 'oldies'

- Models in time domain: ARIMA. Box Jenkins
- Models in frequency domain. Spectral analysis.
- State space models. Series as output of dynamic systems subject to random perturbations.

The Deep Learning tsunami!!!

- Recurrent NN (They are nonlinear AR!!!)
- LSTM-GRU (They are nonlinear AR!!!)
- Transformers and GPTs (They are nonlinear AR!!!)
- AutoResearch

# Advantages of state space models

Natural interpretation through combination of components: trend, seasonal, regression, autoregression

ARMA embedded.

Powerful probabilistic structure facilitating flexible modeling in numerous domains

Recursive computation

Bayesian treatment naturally

Uni and multivariate

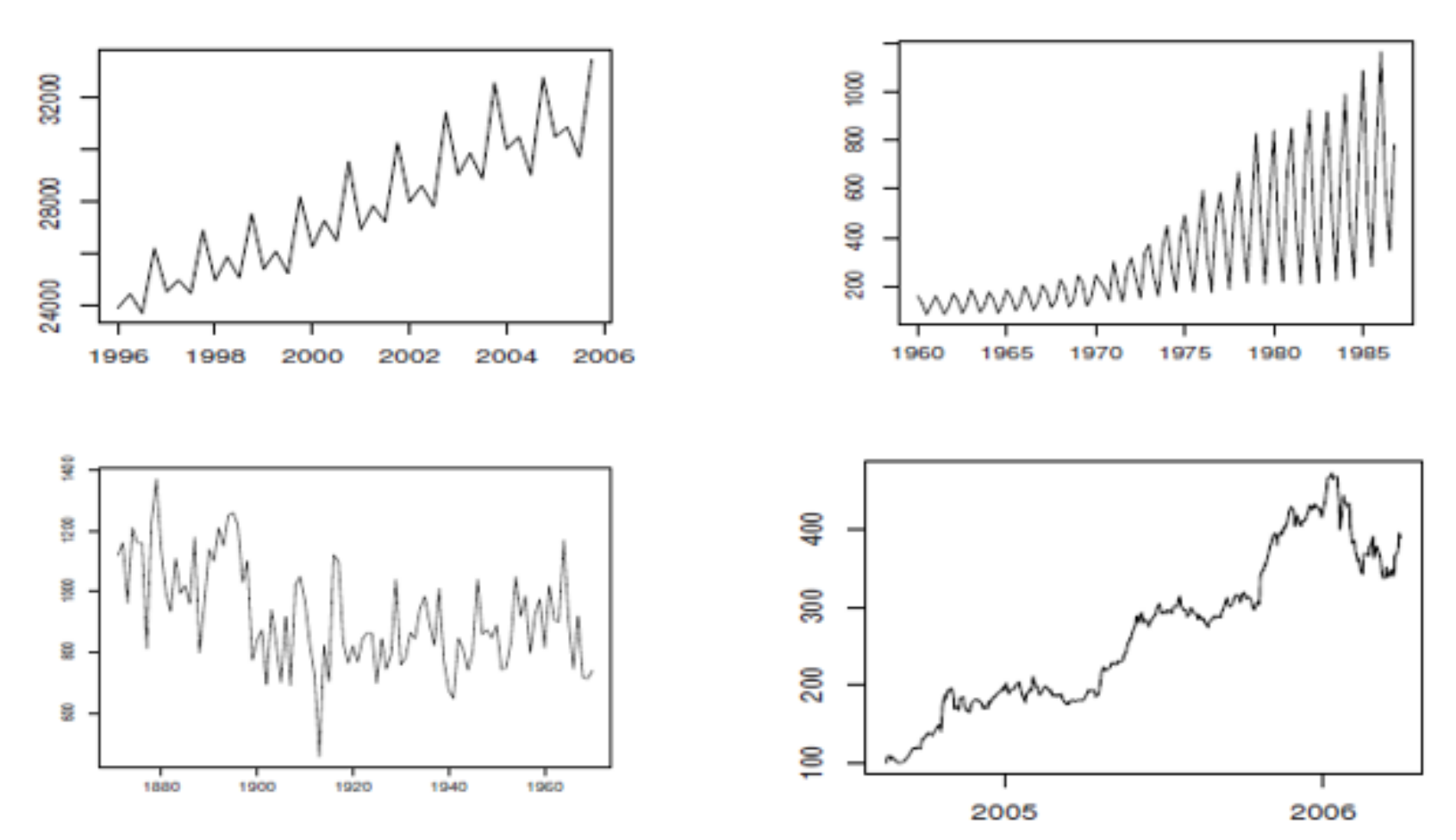
Non-stationarity, structural changes, irregular patterns naturally

Linear (Kalman filter) and non-linear (MCMC).

Very useful in applications

Explainable

# Time series features



# State space models. Intro

# SLAM Example

Object position from measurements subject to random errors

$$\theta \qquad (Y_t : t = 1, 2, \ldots)$$

$$Y_t = \theta + \epsilon_t, \qquad \epsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

$$Y_1, Y_2, \ldots | \theta \overset{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2). \qquad \theta \sim \mathcal{N}(m_0, C_0)$$

$$\theta | y_{1:n} \sim \mathcal{N}(m_n, C_n),$$

$$m_n = \mathrm{E}(\theta | y_{1:n}) = \frac{C_0}{C_0 + \sigma^2/n} \bar{y} + \frac{\sigma^2/n}{C_0 + \sigma^2/n} m_0$$

$$C_n = \mathrm{Var}(\theta | y_{1:n}) = \left( \frac{n}{\sigma^2} + \frac{1}{C_0} \right)^{-1} = \frac{\sigma^2 C_0}{\sigma^2 + n C_0}.$$

Recursively

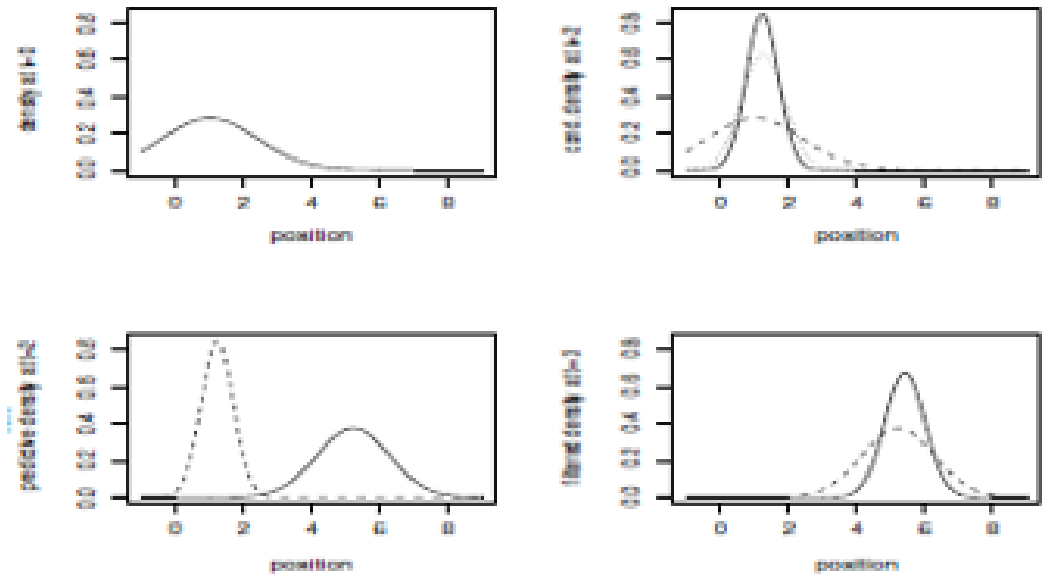
$$m_n = \frac{C_{n-1}}{C_{n-1} + \sigma^2} y_n + \left( 1 - \frac{C_{n-1}}{C_{n-1} + \sigma^2} \right) m_{n-1}$$

$$= m_{n-1} + \frac{C_{n-1}}{C_{n-1} + \sigma^2} (y_n - m_{n-1})$$

$$C_n = \left( \frac{1}{\sigma^2} + \frac{1}{C_{n-1}} \right)^{-1} = \frac{\sigma^2 C_{n-1}}{\sigma^2 + C_{n-1}}.$$

# SLAM Example

| Time | Observation | Mean | Variance |
|---|---|---|---|
| 0 | - | 1 | 2 |
| 1 | 1.3 | 1.24 | 0.4 |
| 2 | 1.2 | 1.222 | 0.222 |
| 3 | ..... | ..... | ..... |



# SLAM Example

- Introduce dynamic component. From t=2, object adopts a velocity

$$\theta_t = \theta_{t-1} + \nu + w_t, \qquad w_t \sim \mathcal{N}(0, \sigma_w^2) \qquad\qquad Y_t = \theta_t + \epsilon_t, \qquad \epsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

With independent noise processes.

- Initial step:

$$\theta_2 | y_{1:2} \sim \mathcal{N}(m_2 = 1.222, C_2 = 0.222).$$

- Prediction:

$$\theta_3 | y_{1:2} \sim \mathcal{N}(a_3, R_3) \qquad\qquad Y_3 | y_{1:2} \sim \mathcal{N}(f_3, Q_3).$$

$$a_3 = \mathrm{E}(\theta_2 + \nu + w_3 | y_{1:2}) = m_2 + \nu = 5.722 \qquad f_3 = \mathrm{E}(\theta_3 + \epsilon_3 | y_{1:2}) = a_3 = 5.722$$

$$R_3 = \mathrm{Var}(\theta_2 + \nu + w_3 | y_{1:2}) = C_2 + \sigma_w^2 = 1.122 \qquad Q_3 = \mathrm{Var}(\theta_3 + \epsilon_3 | y_{1:2}) = R_3 + \sigma^2 = 1.622$$

# SLAM Example

- Initial step: $\theta_2|y_{1:2} \sim \mathcal{N}(m_2 = 1.222, C_2 = 0.222).$
- Forecast: $\theta_3|y_{1:2} \sim \mathcal{N}(a_3, R_3)$ $\quad Y_3|y_{1:2} \sim \mathcal{N}(f_3, Q_3).$

$$a_3 = \mathrm{E}(\theta_2 + \nu + w_3|y_{1:2}) = m_2 + \nu = 5.722 \qquad f_3 = \mathrm{E}(\theta_3 + \epsilon_3|y_{1:2}) = a_3 = 5$$

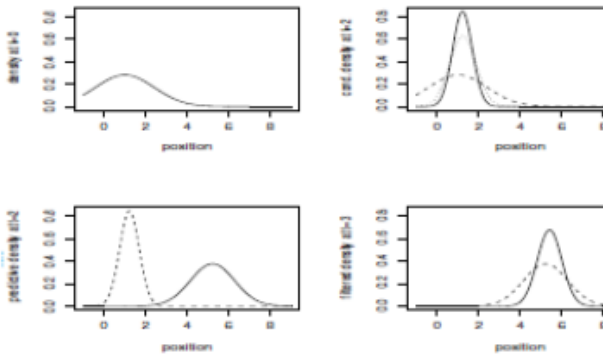
$$R_3 = \mathrm{Var}(\theta_2 + \nu + w_3|y_{1:2}) = C_2 + \sigma_w^2 = 1.122 \qquad Q_3 = \mathrm{Var}(\theta_3 + \epsilon_3|y_{1:2}) = R_3 + \sigma^2$$

- Filter: $Y_3 = 5 \qquad e_t = y_t - f_t = -0.722$

$$\theta_3|y_1, y_2, y_3 \sim \mathcal{N}(m_3, C_3),$$

$$m_3 = a_3 + \frac{R_3}{R_3 + \sigma^2}(y_3 - f_3) = 5.568$$

$$C_3 = \frac{\sigma^2 R_3}{\sigma^2 + R_3} = R_3 - \frac{R_3}{R_3 + \sigma^2} R_3 = 0.346$$



# Lessons learnt

Observable process determined by latent process except for normal error

Latent process depends on previous state linearly, except for normal error

Forecast and estimation sequentially as data arrives

Linearity and normality define dynamic linear models (DLMs)

Time dependence structure, define state space models (SSM)

# SSM definition

- p-variate and m-variate time series $(\theta_t)$ $Y_t$ such that
  - $(\theta_t)$ is Markov chain
  - Given $(\theta_t)$, $Y_t$ are independent and just depends on $\theta_t$

$$\begin{array}{ccccccccccccc}
\theta_0 & \longrightarrow & \theta_1 & \longrightarrow & \theta_2 & \longrightarrow & \cdots & \longrightarrow & \theta_{t-1} & \longrightarrow & \theta_t & \longrightarrow & \theta_{t+1} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \downarrow & & \\
 & & Y_1 & & Y_2 & & & & Y_{t-1} & & Y_t & & Y_{t+1} & &
\end{array}$$

$$\pi(\theta_{0:t}, y_{1:t}) = \pi(\theta_0) \cdot \prod_{j=1}^{t} \pi(\theta_j|\theta_{j-1})\pi(y_j|\theta_j)$$

# DLM definition

- State space model with
  - Observation equation

$$Y_t = F_t\theta_t + v_t, \qquad v_t \sim \mathcal{N}_m(0, V_t)$$

  - State equation

$$\theta_t = G_t\theta_{t-1} + w_t, \qquad w_t \sim \mathcal{N}_p(0, W_t)$$

  - Prior

$$\theta_0 \sim \mathcal{N}_p(m_0, C_0)$$

# DLM definition. Level model

- State space with

$$
\begin{aligned}
Y_t &= \mu_t + v_t, & v_t &\sim \mathcal{N}(0, V) \\
\mu_t &= \mu_{t-1} + w_t, & w_t &\sim \mathcal{N}(0, W)
\end{aligned}
$$

# DLM definition. Linear growth model

- State space with

$$
\begin{aligned}
Y_t &= \mu_t + v_t, & v_t &\sim \mathcal{N}(0, V), \\
\mu_t &= \mu_{t-1} + \beta_{t-1} + w_{t,1}, & w_{t,1} &\sim \mathcal{N}(0, \sigma_\mu^2) \\
\beta_t &= \beta_{t-1} + w_{t,2}, & w_{t,2} &\sim \mathcal{N}(0, \sigma_\beta^2)
\end{aligned}
$$

# SSMs and DLMs: Inference and Prediction

# Key problems

Target $\pi(\theta_s|y_{1:t})$

Filtering $s = t$

Forecasting $s > t$

Smooth $s < t$

$$\begin{array}{ccccccccccccc} \theta_0 & \longrightarrow & \theta_1 & \longrightarrow & \theta_2 & \longrightarrow & \cdots & \longrightarrow & \theta_{t-1} & \longrightarrow & \theta_t & \longrightarrow & \theta_{t+1} & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \downarrow & & \\ & & Y_1 & & Y_2 & & & & Y_{t-1} & & Y_t & & Y_{t+1} & & \end{array}$$

# Prediction with SSMs

- Predictive density for state one step ahead

$$\pi(\theta_t|y_{1:t-1}) = \int \pi(\theta_t|\theta_{t-1})\pi(\theta_{t-1}|y_{1:t-1})\,\mathrm{d}\theta_{t-1}$$

- Predictive density

$$\pi(y_t|y_{1:t-1}) = \int \pi(y_t|\theta_t)\pi(\theta_t|y_{1:t-1})\,\mathrm{d}\theta_t$$

- Filtering density

$$\pi(\theta_t|y_{1:t}) = \frac{\pi(y_t|\theta_t)\pi(\theta_t|y_{1:t-1})}{\pi(y_t|y_{1:t-1})}$$

# Prediction with DLMs (KF)

- Currently, $\theta_{t-1}|y_{1:t-1} \sim \mathcal{N}(m_{t-1}, C_{t-1})$
- State predictive density one step ahead, normal

$$a_t = \mathrm{E}(\theta_t|y_{1:t-1}) = G_t m_{t-1},$$
$$R_t = \mathrm{Var}(\theta_t|y_{1:t-1}) = G_t C_{t-1} G_t' + W_t$$

- Predictive density, normal

$$f_t = \mathrm{E}(Y_t|y_{1:t-1}) = F_t a_t,$$
$$Q_t = \mathrm{Var}(Y_t|y_{1:t-1}) = F_t R_t F_t' + V_t$$

- Filtering density

$$m_t = \mathrm{E}(\theta_t|y_{1:t}) = a_t + R_t F_t' Q_t^{-1} e_t, \qquad e_t = Y_t - f_t$$
$$C_t = \mathrm{Var}(\theta_t|y_{1:t}) = R_t - R_t F_t' Q_t^{-1} F_t R_t$$

# k-steps ahead prediction with SSMs

- Looking ahead in the future

$$\begin{array}{ccccccc} \theta_t & \longrightarrow & \theta_{t+1} & \longrightarrow & \cdots & \longrightarrow & \theta_{t+k} \\ | & & & & & & | \\ Y_{1:t} & & & & & & Y_{t+k} \end{array}$$

- State k steps ahead

$$\pi(\theta_{t+k}|y_{1:t}) = \int \pi(\theta_{t+k}|\theta_{t+k-1})\pi(\theta_{t+k-1}|y_{1:t})\,\mathrm{d}\theta_{t+k-1}$$

- Observation k steps ahead

$$\pi(y_{t+k}|y_{1:t}) = \int \pi(y_{t+k}|\theta_{t+k})\pi(\theta_{t+k}|y_{1:t})\,\mathrm{d}\theta_{t+k}$$

# k-steps ahead prediction with DLMs

- State, normal with mean and variance

$$a_t(k) = G_{t+k} a_{t,k-1},$$
$$R_t(k) = G_{t+k} R_{t,k-1} G'_{t+k} + W_{t+k}$$

- Observation, normal with mean and variance

$$f_t(k) = F_{t+k} a_t(k),$$
$$Q_t(k) = F_{t+k} R_t(k) F'_{t+k} + V_t$$

# Model validation

- Prediction errors

$$e_t = Y_t - \mathrm{E}(Y_t|y_{1:t-1}) = Y_t - f_t \qquad \begin{aligned} e_t &= Y_t - F_t a_t = F_t\theta_t + v_t - F_t a_t \\ &= F_t(\theta_t - a_t) + v_t. \end{aligned}$$

- Properties facilitating validation
  - Expected value is 0
  - Observations from gaussian process
  - $\tilde{e}_t = e_t/\sqrt{Q_t}$ White noise gaussian process

# DLMs: Modeling strategies

# Superposition principle

- The sum of independent DLMs is a DLM

$$Y_t = Y_{1,t} + \cdots + Y_{h,t}$$

$$\begin{aligned} Y_{i,t} &= F_{i,t}\theta_{i,t} + v_{i,t}, & v_{i,t} &\sim \mathcal{N}(0, V_{i,t}), \\ \theta_{i,t} &= G_{i,t}\theta_{i,t-1} + w_{i,t}, & w_{i,t} &\sim \mathcal{N}(0, W_{i,t}) \end{aligned}$$

results in

$$\begin{aligned} Y_t &= F_t\theta_t + v_t, & v_t &\sim \mathcal{N}(0, V_t), \\ \theta_t &= G_t\theta_{t-1} + w_t, & w_t &\sim \mathcal{N}(0, W_t), \end{aligned}$$

$$\theta_t = \begin{bmatrix} \theta_{1,t} \\ \vdots \\ \theta_{h,t} \end{bmatrix} \qquad F_t = [F_{1,t}|\cdots|F_{h,t}] \qquad G_t = \begin{bmatrix} G_{1,t} & & \\ & \ddots & \\ & & G_{h,t} \end{bmatrix} \qquad W_t = \begin{bmatrix} W_{1,t} & & \\ & \ddots & \\ & & W_{h,t} \end{bmatrix} \qquad V_t = \sum_{i=1}^{j} V_{i,t}$$

Model building strategy based on blocks

# Polynomial trend models

- Order n

$$\begin{cases} Y_t = \theta_{t,1} + v_t \\ \theta_{t,j} = \theta_{t-1,j} + \theta_{t-1,j+1} + w_{t,j} \quad j = 1, \ldots, n-1 \\ \theta_{t,n} = \theta_{t-1,n} + w_{t,n}. \end{cases}$$

$$F = (1, 0, \ldots, 0)$$

$$G = \begin{bmatrix} 1 & 1 & 0 & & \ldots & 0 \\ 0 & 1 & 1 & 0 & \ldots & 0 \\ \vdots & & & & \ddots & \vdots \\ 0 & & \ldots & 0 & 1 & 1 \\ 0 & & \ldots & & 0 & 1 \end{bmatrix}$$

$$W = \mathrm{diag}(W_1, \ldots, W_n).$$

- Predictive function

$$f_t(k) = E(Y_{t+k}|y_{1:t}) = a_{t,0} + a_{t,1}k + \cdots + a_{t,n-1}k^{n-1}$$

- Consider cases n=1,2,3

# Seasonal models

- State vector dimension s-1

$$F = (1, 0, \ldots, 0)$$

$$G = \begin{bmatrix} -1 & -1 & \ldots & -1 & -1 \\ 1 & 0 & & 0 & 0 \\ 0 & 1 & & 0 & 0 \\ & & \ddots & & \\ 0 & 0 & & 1 & 0 \end{bmatrix}.$$

NB: Better results with Fourier decomposition (retain only meaningful components)

# (Dynamic) regression models

- From the standard regression model

$$Y_t = \beta_1 + \beta_2 x_t + \epsilon_t, \quad \epsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

- To the dynamic regression model

$$Y_t = \beta_{1,t} + \beta_{2,t} x_t + \epsilon_t, \quad \epsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

$$\begin{aligned} Y_t &= x_t' \theta_t + v_t, & v_t &\sim \mathcal{N}(0, \sigma_t^2) \\ \theta_t &= G_t \theta_{t-1} + w_t, & w_t &\sim \mathcal{N}_p(0, W_t) \end{aligned}$$

# Modeling strategy

- Trend + Seasonal + Regression term+ Low order AR term (1)

$\theta_t$

```
dlmModPoly(2) + dlmModSeas(4)
```

# Final comments

# Final comments

- Again we have gone through the 'easy' models.
- $U$ or $V$ unknown?
- Non-normal?
- Non-linear? e.g. NNs: Transformers, GPTs,...

# References

# References

West, Harrison, Bayesian forecasting and dynamic models, Springer
Prado, West Time series: modeling, computation, and inference
Petris, Petrone, Campagnoli. Dynamic linear models

Murphy, Probabilistic Machine Learning
Menchero, Diez, Insua, Müller, Bayesian analysis of nonlinear autoregression models based on neural networks. Neural computation
PML Reading group Check Session 5

# Lab 5

# Lab5

DLMS in R with dlm